

1. Filter for DNS Traffic

Use the display filter: dns keyword

dns : lọc ra dns packets

dns.qry.name.len>30 : lọc ra DNS query có độ dài lớn hơn 30 bytes

!(dns.qry.name contains "mozilla") and !(.dns.qry.name contains "google") and !(dns.qry.name contains "microsoft") and !(dns.qry.name contains "ubuntu") and !(dns.qry.name contains "facebook") : loại bỏ các domain không quan trọng

The image shows a Wireshark capture of network traffic. The filter bar at the top is set to 'dns'. The packet list on the left shows several DNS packets. The packet details pane on the right shows the structure of a DNS query, including the question section with the domain '8.8.8.8'.

No.	Time	Source	Destination	Protocol	Length	Time to Live	Window	Syn	Info
14837	21:35:39.957033	192.168.30.11	8.8.8.8	DNS	107	64	64240	No.	Standard query 0xe95e A 0
14917	21:35:40.004354	8.8.8.8	192.168.30.11	DNS	203	127	64240	No.	Standard query response 0x
20173	21:36:03.398560	192.168.20.11	1.1.1.1	DNS	105	64			Standard query 0xa3ba A 1r
20174	21:36:03.398589	192.168.20.11	1.1.1.1	DNS	105	64			Standard query 0xe69c AAAA
20233	21:36:03.433805	1.1.1.1	192.168.20.11	DNS	180	127			Standard query response 0x
20234	21:36:03.449347	1.1.1.1	192.168.20.11	DNS	243	127			Standard query response 0x
20235	21:36:03.450272	192.168.20.11	1.1.1.1	DNS	120	64			Standard query 0x85be AAAA
20236	21:36:03.490505	1.1.1.1	192.168.20.11	DNS	202	127			Standard query response 0x
20247	21:36:03.729309	192.168.20.11	1.1.1.1	DNS	99	64			Standard query 0x572f A de
20249	21:36:03.729411	192.168.20.11	1.1.1.1	DNS	99	64			Standard query 0x5c69 AAAA
20257	21:36:03.775998	1.1.1.1	192.168.20.11	DNS	210	127			Standard query response 0x
20258	21:36:03.781746	1.1.1.1	192.168.20.11	DNS	222	127			Standard query response 0x
20261	21:36:03.792594	192.168.20.11	1.1.1.1	DNS	88	64			Standard query 0xf035 AAAA
20269	21:36:03.829678	1.1.1.1	192.168.20.11	DNS	145	127			Standard query response 0x
20270	21:36:03.830778	192.168.20.11	1.1.1.1	DNS	88	64			Standard query 0x5a66 AAAA
20273	21:36:03.877130	1.1.1.1	192.168.20.11	DNS	145	127			Standard query response 0x
20300	21:36:07.672924	192.168.30.11	8.8.8.8	DNS	94	64	64240	No.	Standard query 0x466f A af
20301	21:36:07.673043	192.168.30.11	8.8.8.8	DNS	94	64	64240	No.	Standard query 0x1165 AAAA
20304	21:36:07.734806	8.8.8.8	192.168.30.11	DNS	198	127	64240	No.	Standard query response 0x

The image shows a Wireshark capture of network traffic, filtered by 'dns.qry.name.len>30'. The packet list on the left shows several DNS queries. The packet details pane on the right shows the structure of a DNS query, including the question section with the domain '8.8.8.8'.

No.	Time	Source	Destination	Protocol	Length	Time to Live	Window	Syn	Info
20235	21:36:03.450272	192.168.20.11	1.1.1.1	DNS	120	64			Standard query 0x85be AAAA
20236	21:36:03.490505	1.1.1.1	192.168.20.11	DNS	202	127			Standard query response 0x
20493	21:36:13.111884	192.168.20.11	1.1.1.1	DNS	120	64			Standard query 0x5a66 AAAA
20498	21:36:13.153081	1.1.1.1	192.168.20.11	DNS	202	127			Standard query response 0x
20512	21:36:13.254968	192.168.20.11	1.1.1.1	DNS	112	64			Standard query 0xa3d7 A f
20513	21:36:13.255011	192.168.20.11	1.1.1.1	DNS	112	64			Standard query 0xda58 AAAA
20522	21:36:13.293287	1.1.1.1	192.168.20.11	DNS	260	127			Standard query response 0x
20523	21:36:13.296133	1.1.1.1	192.168.20.11	DNS	212	127			Standard query response 0x
22550	21:36:53.067133	192.168.20.11	1.1.1.1	DNS	120	64			Standard query 0x47cd AAAA
22561	21:36:53.114037	1.1.1.1	192.168.20.11	DNS	202	127			Standard query response 0x
22643	21:36:53.456133	192.168.20.11	1.1.1.1	DNS	120	64			Standard query 0xe4be AAAA
22692	21:36:53.499274	1.1.1.1	192.168.20.11	DNS	202	127			Standard query response 0x
23739	21:37:45.654033	192.168.30.11	8.8.8.8	DNS	137	64	63903	No.	Standard query 0xb1ae AAAA
23741	21:37:45.714695	8.8.8.8	192.168.30.11	DNS	230	127	64240	No.	Standard query response 0x
23773	21:37:45.984772	192.168.30.11	8.8.8.8	DNS	114	64	63903	No.	Standard query 0x114b A c
23774	21:37:45.984818	192.168.30.11	8.8.8.8	DNS	114	64	63903	No.	Standard query 0xcba0 AAAA
23777	21:37:46.047582	8.8.8.8	192.168.30.11	DNS	214	127	64240	No.	Standard query response 0x
23779	21:37:46.107898	8.8.8.8	192.168.30.11	DNS	262	127	64240	No.	Standard query response 0x
23806	21:37:46.592859	192.168.30.11	8.8.8.8	DNS	115	64	63903	No.	Standard query 0xb7b0 A f

Phát hiện các subdomain lạ đến từ cùng 1 domain smartbin.live và host truy vấn

network_traffic2.pcapng																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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[dns.qry.name.len>30 and (dns.qry.name contains "mozilla") and !(dns.qry.name contains "google") and !(dns.qry.name contains "microsoft") and !(dns.qry.name contains "ubuntu") and !(dns.qry.name contains "facebook")]																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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22..	21:50:11...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x72e7	CNAME 49a00164c07919bd04dbf1001d1e72473f.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:11...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x41ea	TXT 344f015011db16eafb3a310038c670da57.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:12...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x60e0	MX 01200164c0a7a313b533eb001e9e742fab.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:12...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x01cf	TXT 34a801501135f73ef29f5900394d82f255.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:13...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x4f5b	TXT 07d40164c0519aac4eda1001f0e429832.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:14...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0xa121	MX 02c40150115b02e4bbefbd003a3d64914d.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:14...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x164c	CNAME 200b0164c09638f4f0a4690020f1921f1f.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:15...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x4ff1	TXT 29d0015011a1e81a3f7bd003bd098f1b4.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:15...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x078d	MX 55e00164c0c88e1eb59aec0021dc77a464.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:16...	192.168.10.2	8.8.8.8	DNS	138	128		Standard query 0x6c99	TXT 61b80164c06a70500d6f4a0022e6b7bf93bf055d8523db54007bcd379947.sm...																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:16...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x296a	MX 5a390150114a41de1f8514003c6fc4e484.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:16...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x2dc4	CNAME 6d010164c07fa8b916e12b0023cab2df0f.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:17...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x64d1	CNAME 1de20150110873695133e003d9f0c70bd.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:17...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x0e9d	CNAME 2f8a0164c06b129dc648d70024081fe7b4.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:18...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x07c7	CNAME 5737015011008f373d4c4a003e1ff977cd.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:18...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x5a1e	TXT 239b0164c017a916041a3100250e59dca0.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:19...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x6cf5	TXT 22250150118411ee1ec7bc003f81be7efc.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:19...	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x1fca	CNAME 44d70164c0e97b3d5388ae002677c9d2af.smartbin.live																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:20...	192.168.10.2	8.8.8.8	DNS	297	128		Standard query 0x3f61	MX 5d6a0164c030963cc58ee0027fb1788f9a21ae8d4174a69741eb5905591.cbf...																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
22..	21:50:20...	192.168.10.2	8.8.8.8	DNS	297	128		Standard query 0x1b2e	MX 60df0164c096d923f73a350028f6ca319503f123346e7432b394ef0df7f.d17...																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
> Frame 217233: 112 bytes on wire (896 bits), 112 bytes captured (896 bits) on interf...						0000	50	39	93	00	05	01	ef	4b	25	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

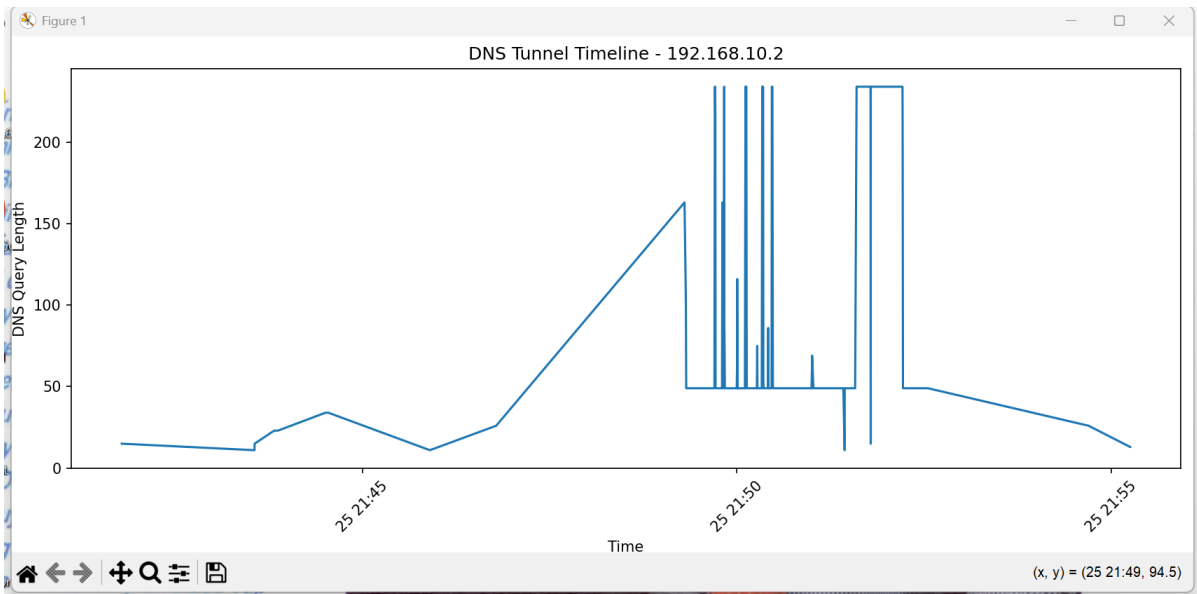
Lọc ra domain nghi ngờ

network_traffic2.pcapng									
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help									
[dns and dns.qry.name.len>30 and (dns.qry.name contains "smartbin") and ip.src == 192.168.10.2]									
No.	Time	Source	Destination	Protocol	Length	Time to Live	Window	Info	
217218	21:49:18.002692	192.168.10.2	8.8.8.8	DNS	226	128		Standard query 0x1cfd TXT 21b603501100000001065cda2f947feacd7185bd9c3fa67f...	
217229	21:49:19.091722	192.168.10.2	8.8.8.8	DNS	165	128		Standard query 0x18ca TXT 05c3005011855904a7a36f0000b1e4d66d91fa45cf0b9d241...	
217233	21:49:19.358187	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x3f61 MX 527b015011b5098c4b97600018fd7c9cc.smartbin.live	
217240	21:49:20.387745	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x274b MX 0b3d015011488725516a5600026b15df56.smartbin.live	
217243	21:49:21.441843	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x5da3 CNAME 52500150112a3e6502d2d200032cb8ca5e.smartbin.live	
217807	21:49:22.478388	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x700b CNAME 77db015011c76143f0d99300040b9dd8e7.smartbin.live	
217810	21:49:23.574131	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x7ffb MX 2aff0150110d53a1bd57d30005ca752360.smartbin.live	
217841	21:49:24.598397	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x3fbd MX 378601501174bb82f8fb880006eb71d6ad.smartbin.live	
217939	21:49:25.642588	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x0cb1 TXT 27000150112b9b390adff0007a7a5c6b6.smartbin.live	
217963	21:49:26.660184	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x4e32 MX 3193015011d1bb422f71ef0000f0d05c89.smartbin.live	
217989	21:49:27.694938	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x32cc TXT 7d130150119b363d542c3200099050e631.smartbin.live	
218013	21:49:28.730634	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x5ca8 MX 1020015011ce9e3fb6a5c8000a288b565e.smartbin.live	
218052	21:49:29.748420	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x3f8a CNAME 3996015011b45dd56fa9f5000b4368afb.smartbin.live	
218388	21:49:30.816678	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x2826 MX 7c7d01501180b3055ac885000cad31124.smartbin.live	
218398	21:49:31.854180	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x1002 CNAME 0a79015011984842f5a788000d86a26b32.smartbin.live	
218419	21:49:32.937945	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x60bc CNAME 208c015011ad0aaec6507d000edac964a.smartbin.live	
218421	21:49:33.989858	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x6949 CNAME 1fb0015011da3d799b0e1e2000f14da41ab.smartbin.live	
218431	21:49:35.027641	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0xb616 MX 4b6f015011bbe8dd13d7f90010592d11f0.smartbin.live	
> Frame 217218: 226 bytes on wire (1808 bits), 226 bytes captured (1808 bits) on interface 0 > Ethernet II, Src: 50:ef:b5:00:01:00 (50:ef:b5:00:01:00), Dst: 50:39:93:00:05:01 (50:39:93:00:05:01) 0 > 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 10 0 > Internet Protocol Version 4, Src: 192.168.10.2, Dst: 8.8.8.8 0 > User Datagram Protocol, Src Port: 57599, Dst Port: 53 0 Domain Name System (query) 0 Transaction ID: 0x1cfd 0 Flags: 0x0100 Standard query 0 Questions: 1 0 Answer RRs: 0 0									

Chọn follow stream để theo dõi trình tự các gói tin request và response

No.	Time	Source	Destination	Protocol	Length	Time to Live	Window	Info
217218	21:49:18.00			WNS	226	128		Standard query 0x1cfd TXT 21b603501100000001065cda2f947feacd7185bd9c3fa67f...
217224	21:49:18.4	Mark/Unmark Selected	Ctrl+M	WNS	385	127		Standard query response 0x1cfd TXT 21b603501100000001065cda2f947feacd7185b...
217229	21:49:19.0	Ignore/Unignore Selected	Ctrl+D	WNS	165	128		Standard query 0x18ca TXT 05c3005011855904a7a36f0000b1e4d66d91fa45cf0b9d241...
217231	21:49:19.3	Set/Unset Time Reference	Ctrl+T	WNS	212	127		Standard query response 0x18ca TXT 05c3005011855904a7a36f0000b1e4d66d91fa45...
217233	21:49:19.3	Time Shift...		WNS	112	128		Standard query 0x3f61 MX 527b0150111b5098c4b97600018fd7c9cc.smartbin.live
217234	21:49:19.6	Packet Comments	Ctrl+Shift+T	WNS	163	127		Standard query response 0x3f61 MX 527b0150111b5098c4b97600018fd7c9cc.smartb...
217240	21:49:20.3			WNS	112	128		Standard query 0x274b MX 0b3d01501148872551ba5600026b15df56.smartbin.live
217241	21:49:20.5	Edit Resolved Name		WNS	163	127		Standard query response 0x274b MX 0b3d01501148872551ba5600026b15df56.smartb...
217243	21:49:21.4			WNS	112	128		Standard query 0x5da3 CNAME 52500150112a3e6502d2d200032cb8ca5e.smartbin.live
217244	21:49:21.5	Apply as Filter		WNS	161	127		Standard query response 0x5da3 CNAME 52500150112a3e6502d2d200032cb8ca5e.sma...
217807	21:49:22.4	Prepare as Filter		WNS	112	128		Standard query 0x700b CNAME 77db015011c76143f0d99300040b9dd8e7.smartbin.live
217808	21:49:22.5	Conversation Filter		WNS	161	127		Standard query response 0x700b CNAME 77db015011c76143f0d99300040b9dd8e7.sma...
217810	21:49:23.5	Colorize Conversation		WNS	112	128		Standard query 0x7ffb MX 2aff0150110d53a1bd57d30005ca752360.smartbin.live
217811	21:49:23.7	SCTP		WNS	163	127		Standard query response 0x7ffb MX 2aff0150110d53a1bd57d30005ca752360.smartb...
217841	21:49:24.5	Follow		WNS	112	128		Standard query 0x3fbd MX 378601501174bb82f8fb880006b71d6ad.smartbin.live
217844	21:49:24.8			WNS	385	127		Standard query response 0x3fbd MX 378601501174bb82f8fb880006b71d6ad.smartb...
217939	21:49:25.6			WNS	159	127		Standard query 0x0cb1 TXT 27000150112b9b390adff0007a7a5c6b6.smartbin.live
217953	21:49:25.7	Copy		WNS	159	127		Standard query response 0x0cb1 TXT 27000150112b9b390adff0007a7a5c6b6.smart...

2. Identify Tunneling Indicators



Biểu đồ timeline thể hiện dns packets từ host 192.168.10.2

⇒ Dễ thấy tần suất cũng như độ dài gói tin DNS tăng mạnh trong 1 khoảng thời gian nhất định

Gói tin truy vấn DNS đầu tiên từ 192.168.10.2 đến subdomain

Loại record: TXT

Đối với 1 host bình thường thì việc truy vấn TXT record cực kỳ hiếm xảy ra

network_traffic2.pcapng

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udpstream eq 657

No.	Time	Source	Destination	Protocol	Length	Time to Live	Window	Info
217218	21:49:18.002692	192.168.10.2	8.8.8.8	DNS	226	128		Standard query 0x1cfd TXT 21b603501100000001065cda2f947feacd7185bd9c3fa67fc1
217224	21:49:18.427863	8.8.8.8	192.168.10.2	DNS	385	127		Standard query response 0x1cfd TXT 21b603501100000001065cda2f947feacd7185bd9
217229	21:49:19.091722	192.168.10.2	8.8.8.8	DNS	165	128		Standard query 0x18ca TXT 05c3005011855904a7a36f0000b1e4d66d91fa45cf0b9d241af
217231	21:49:19.356534	8.8.8.8	192.168.10.2	DNS	212	127		Standard query response 0x18ca TXT 05c3005011855904a7a36f0000b1e4d66d91fa45cf
217233	21:49:19.358187	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x3f61 MX 527b0150111b5098c4b97600018fd7c9cc.smartbin.live
217234	21:49:19.632125	8.8.8.8	192.168.10.2	DNS	163	127		Standard query response 0x3f61 MX 527b0150111b5098c4b97600018fd7c9cc.smartbin
217240	21:49:20.387745	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x274b MX 0b3d01501148872551ba5600026b15df56.smartbin.live
217241	21:49:20.558053	8.8.8.8	192.168.10.2	DNS	163	127		Standard query response 0x274b MX 0b3d01501148872551ba5600026b15df56.smartbin

0... .. = Response: Message is a query
.000 0... .. = Opcode: Standard query (0)
... ..0... .. = Truncated: Message is not truncated
... ..1... .. = Recursion desired: Do query recursively
... ..0... .. = Z: reserved (0)
... ..0... .. = Non-authenticated data: Unacceptable

Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0

Queries

- 21b603501100000001065cda2f947feacd7185bd9c3fa67fc15abca9867.6999511a8810f5206498c4aba151378bfd9cdd4950283396281af78aec9d.6bb0a2237ab10d69b06b35d4c8.smartbin.live: type: TXT (16) (Text strings)
[Name Length: 162]
[Label Count: 5]
[Type: TXT (16) (Text strings)]
[Class: IN (0x0001)]
[Response In: 217224]

Và trong DNS tunneling: TXT records thường bị lạm dụng để malware gửi dữ liệu qua DNS. Sau đó, Server phản hồi TXT record và Malware giải mã để nhận lệnh.

DNS response

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udpstream eq 657

No.	Time	Source	Destination	Protocol	Length	Time to Live	Window	Info
217218	21:49:18.002692	192.168.10.2	8.8.8.8	DNS	226	128		Standard query 0x1cfd TXT 21b603501100000001065cda2f947feacd7185bd9c3fa67fc1
217224	21:49:18.427863	8.8.8.8	192.168.10.2	DNS	385	127		Standard query response 0x1cfd TXT 21b603501100000001065cda2f947feacd7185bd9
217229	21:49:19.091722	192.168.10.2	8.8.8.8	DNS	165	128		Standard query 0x18ca TXT 05c3005011855904a7a36f0000b1e4d66d91fa45cf0b9d241af
217231	21:49:19.356534	8.8.8.8	192.168.10.2	DNS	212	127		Standard query response 0x18ca TXT 05c3005011855904a7a36f0000b1e4d66d91fa45cf
217233	21:49:19.358187	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x3f61 MX 527b0150111b5098c4b97600018fd7c9cc.smartbin.live
217234	21:49:19.632125	8.8.8.8	192.168.10.2	DNS	163	127		Standard query response 0x3f61 MX 527b0150111b5098c4b97600018fd7c9cc.smartbin
217240	21:49:20.387745	192.168.10.2	8.8.8.8	DNS	112	128		Standard query 0x274b MX 0b3d01501148872551ba5600026b15df56.smartbin.live
217241	21:49:20.558053	8.8.8.8	192.168.10.2	DNS	163	127		Standard query response 0x274b MX 0b3d01501148872551ba5600026b15df56.smartbin

.000 0... .. = Opcode: Standard query (0)
... ..0... .. = Authoritative: Server is not an authority for domain
... ..0... .. = Truncated: Message is not truncated
... ..1... .. = Recursion desired: Do query recursively
... ..1... .. = Recursion available: Server can do recursive queries
... ..0... .. = Z: reserved (0)
... ..0... .. = Answer authenticated: Answer/authority portion was not authenticated by the server
... ..0... .. = Non-authenticated data: Unacceptable
... ..0000 = Reply code: No error (0)

Questions: 1
Answer RRs: 1
Authority RRs: 0
Additional RRs: 0
Queries

- 21b603501100000001065cda2f947feacd7185bd9c3fa67fc15abca9867.6999511a8810f5206498c4aba151378bfd9cdd4950283396281af78aec9d.6bb0a2237ab10d69b06b35d4c8.smartbin.live: type: TXT, request In: 217218
Time: 0.425171000 seconds]

Tương tự như vậy cho các bản tin sau.

Kết luận : Dựa vào dữ liệu trên, host 192.168.10.2 có khả năng đã bị malware tấn công và sử dụng DNS tunneling để chuyển dữ liệu nhằm che dấu hoạt động độc hại.